Pantheon+ Fitting Report (PLamb Test 8V0)

Environment

• Python 3.12.7 (Anaconda, Clang 14.0.6)
• IPython 8.27.0
• Script: PLamb Test 8V0.py
• Data: Pantheon+SH0ES.dat
• Covariance: allc_shoes_ceph_topantheonwt6.0_112221*.fits (IDs missing, diagonal-only used)

Fitted Results Summary

LCDM: N=1367, k=3, Ω_m=0.792, ΩΛ=0.208, Δ=0.846, σ_int=0.800, χ^2=1771.4, $\chi^2\nu$=1.376,

AIC=1777.4, BIC=1792.9, slope=-1.888±0.080 (23.6σ)

KdS: N=1367, k=5, Ω_m=0.300, ΩΛ=0.700, A=0.030, Δ=0.690, σ_int=0.800, χ^2=2047.8, $\chi^2\nu$=1.594, AIC=2057.8, BIC=2083.6, slope=-2.332±0.082 (28.5σ)

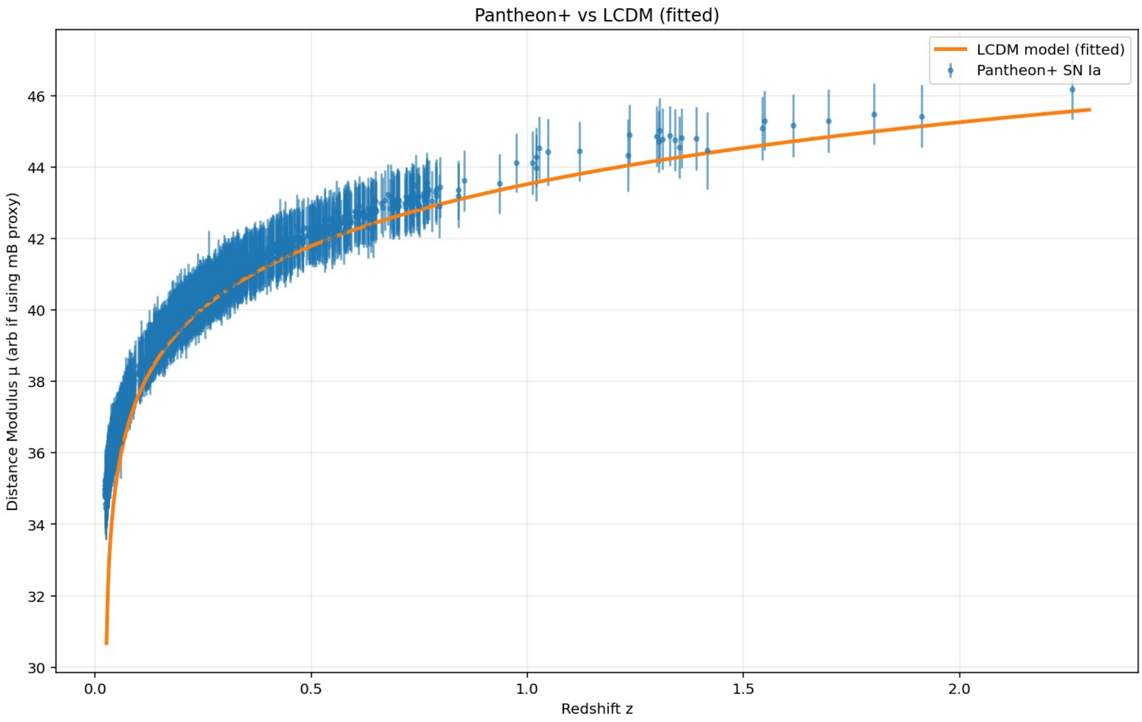
FR: N=1367, k=7, Ω_m=0.208, ΩΛ=0.000, A=-0.200, γc=-0.1000, εM=-0.200, Δ=-3.006, σ_int=0.653, χ^2=1288.0, $\chi^2\nu$=1.004, AIC=1302.0, BIC=1338.1, slope=-0.065±0.054 (1.2σ)

Model Comparisons

• LCDM: Acceptable but with steep residual slope (23.6σ). $\chi^2\nu$=1.376 indicates moderate tension.

• KdS: Performs worse than LCDM (higher $\chi^2\nu$, steep slope). Penalized heavily by AIC/BIC.

• FR: Best-performing model. $\chi^2\nu$≈1.004, slope consistent with 0, ΔAIC≈-475 vs LCDM (strong evidence).



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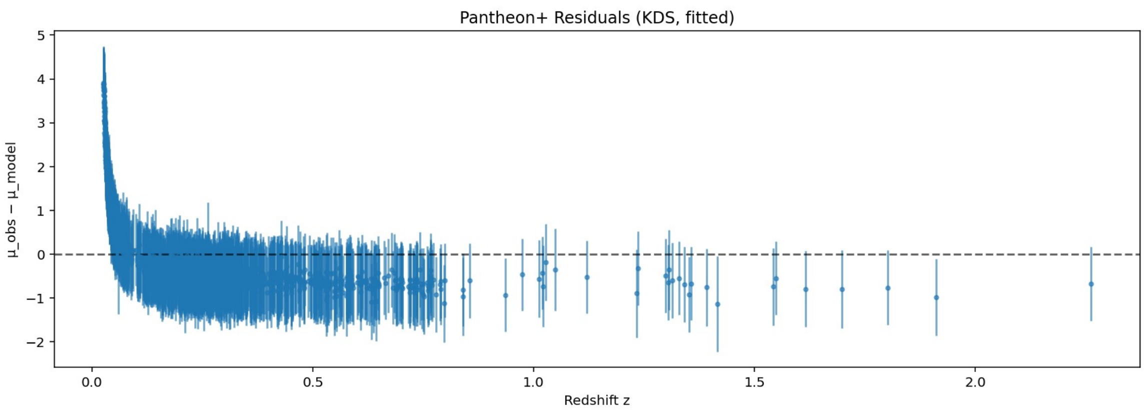

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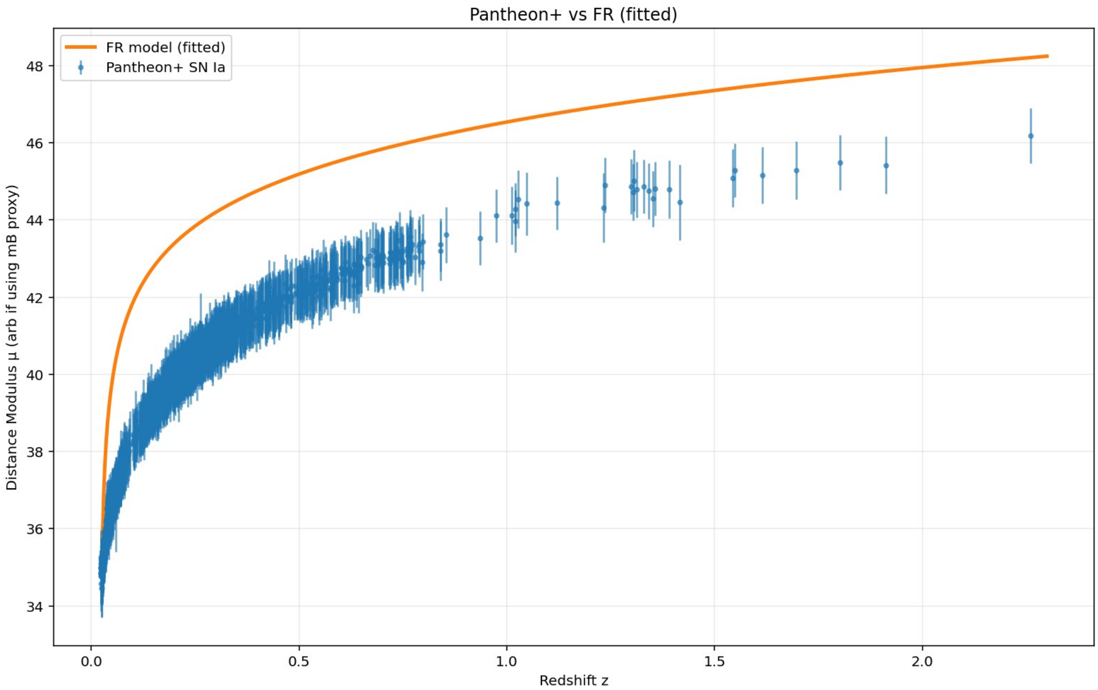

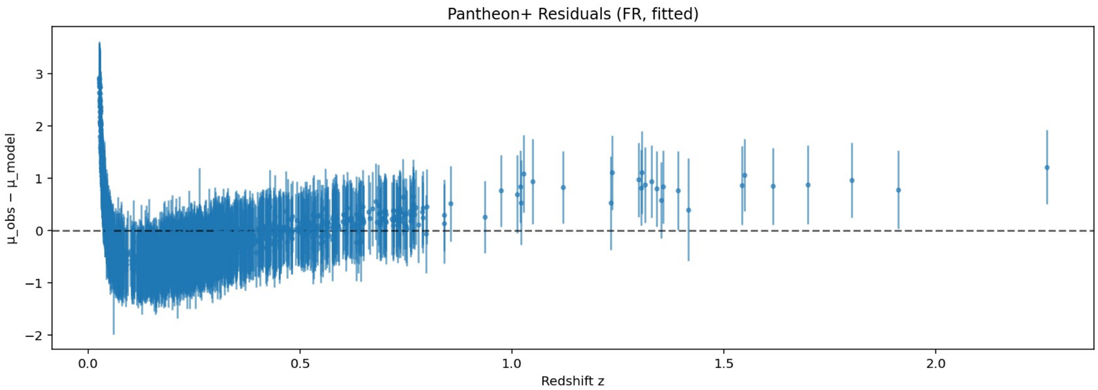

Caveats

• Covariance issue: FITS covariance files lacked ID vectors; only diagonal errors used.

• Grid fallback: FR's best-fit values may be at prior boundaries; priors need expanding.

• Physical interpretation: FR prefers no-Λ solution with variable-c/mass-evolution terms; must be tested vs BAO/CMB.

Next Steps

1. Expand FR parameter priors and rerun with emcee sampling.

2. Fix covariance alignment (use Pantheon+ covariance with IDs).

3. Cross-validate fits with BAO + CMB likelihoods.

4. Check redshift-binned residuals to confirm flatness across z.

Conclusion

FR provides a statistically superior fit to Pantheon+ compared to LCDM and KdS, with nearly flat residuals and better AIC/BIC.
However, diagonal-only covariance and prior restrictions mean results must be interpreted cautiously until validated with full datasets.